

alludes to the behaviour of these tiny creatures that follow others blindly, eventually jumping off a cliff. The phrase typically denotes a person who unthinkingly joins a mass movement, especially a headlong rush to destruction. Unless moderators know the meaning of the specific words, they cannot correctly classify the post as offensive or not. Hence, automated techniques that use external knowledge for linking entities, events and mentions in textual content can provide additional information to human moderators to aid in their classification task.

Also, content for moderation can contain a mix of languages, often known as code-mixing. This again causes problems for human moderators as they may not be familiar with multiple languages. Also, the cultural and social context implicitly assumed in the social media post may not match that of the human moderator. Hence, it is important to make sure that content for moderation is appropriately assigned to a moderator with the right skills to analyse it. Given the profiles of human moderators, the incoming textual content can be analysed by NLP techniques to route it to a moderator who is best suited for that content. While automated techniques may not completely replace the human moderator, they can definitely help in aiding the human moderator in many ways such as those mentioned above, by providing additional context to difficult and unknown terms, entities and events, as well as helping to route the content to the most appropriately skilled moderator, based on the content.

In last month's column, I had mentioned the use of lexicons containing a list of abusive terms, in offensive content detection. These lexicons can be directly used in keyword spotting systems or their presence/absence can serve as features to a machine learning based classifier. However, usage of such 'offensive' words in machine learning approaches comes with its own issues. Given that social media language is rapidly evolving over time, new slurs or offensive terms get introduced. More importantly, people who create offensive content try to elude content moderation checks by deliberately creating code words for abusive language which are not part of any 'offensive content' lexicon.

Consider the following tweet: "Gas the skypes." This is an offensive tweet since 'skypes' is a derogatory code word for Jews. On the other hand, the post, "My husband skypes his mom every morning" is not an offensive content as the word 'skype' is used in the context of making an Internet phone call using the software product Skype. Code words such as 'skypes', 'skittles', 'yahoo', etc, are getting dynamically created by those who post offensive content online. Hence any static 'offensive content' lexicon will not be able to keep up with such evolving slurs.

This highlights an important challenge for any content moderation system: Is it possible to surface out such offensive code words automatically from the online content? The key to attacking this problem lies in understanding the context in which the offensive code word is used. Consider the examples

we discussed earlier. The post, "My husband always skypes his mom every morning," is used in the contextual frame of communication. On the other hand, the tweet, "Gas the skypes," has the contextual frame where 'skypes' stands for a specific group of people. This contextual difference can be used to identify code words, but it is still left to the human moderator to make the decision on whether the code word encountered is offensive or not.

Most of the current research in content moderation systems is focused around improving the performance of automated approaches. It fails to address the fact that we are still a long way from a fully automatic approach and, hence, hybrid systems hold the key to scalable and accurate online content moderation on social media sites. In the case of hybrid systems where automated techniques co-exist with manual moderators, it is essential to design the automated content moderation techniques not just for better classification accuracy, but also to enable them to aid the human moderators. Natural language processing can be applied to help provide more contextual and external knowledge to human moderators, enabling them to achieve greater productivity as well as improving their classification accuracy.

While there are a number of data sets that are being made available for content moderation, what characterises many of these is that the positive instances for offensive content have typically been collected using specific 'key words' which can denote offensive content. This can lead to a bias in the data set, and this in turn can prevent generalisation of the model on real-world social media data. Given that the number of samples for offensive content is generally a very small percentage of the overall content, it is not possible to collect a general corpus and annotate it to get a significant number of positive samples, which can be used for supervised training. While most of the current approaches for classifying offensive content on social media have been based on supervised learning techniques, the lack of large volumes of annotated data means that we need to turn to unsupervised techniques. We will cover this in next month's column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! 

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